# Neev Rao

512-662-5593 | neev2024@gmail.com | Austin, TX
github.com/neev-rao | linkedin.com/in/neev-rao-97a772250

## EDUCATION

**The University of Texas at Austin**, Austin, TX — Aug 2024 - May 2028
*Bachelor of Science, Computer Science*

**GPA:** 3.81

**Relevant Coursework:** Data Structures, Algorithms, Operating Systems, Computer Architecture, Network Security & Privacy, Machine Learning, Object-Oriented Programming, Software Engineering, Symbolic Programming

## SKILLS

**Languages:** Python, C++, C, Java, JavaScript, HTML/CSS
**Frameworks:** Spring Boot, React, Node.js, Express.js, Bootstrap
**Tools & Technologies:** SQL, Git, Docker, GitHub Copilot, Devin Desktop, CI/CD, Linux, Kubernetes

## EXPERIENCE

**Dell**, Austin, TX — May 2026 - Aug 2026
*Software Engineering Intern*

- Designed a containerized, cluster-level telemetry agent in Go to monitor compute nodes running intensive AI workloads, utilizing spec-driven development with Devin Desktop to accelerate architecture and deployment.
- Developed custom Kubernetes API collectors to continuously aggregate cluster-level health and resource metrics, routing real-time observability data to the central Dell Automation Platform Orchestrator.
- Implemented a secure, high-throughput NATS messaging pipeline, including custom data envelopes, JSON marshaling, and CBOR signing to validate payloads over mTLS-encrypted connections.

**Trend Micro**, Austin, TX — Jun 2025 - Aug 2025
*Software Research & Development Intern*

- Developed a scalable, multi-container Threat Protection System emulator backed by a MariaDB database to simulate large-scale customer configurations for internal testing, reducing reliance on costly physical hardware.
- Designed and implemented a Java Spring Boot application deployed in isolated Docker containers.
- Integrated SOAP web services to communicate with the management system, ensuring existing compatibility.

**Mercor**, Remote — Feb 2025 - Apr 2025
*Contractor, Machine Learning Optimization*

- Collaborated with a leading AI company to fine-tune LLMs for image recognition tasks.
- Created expert solutions to complex computer science and mathematics problems for AI model evaluation.
- Evaluated AI-generated responses against human solutions to identify and correct flaws.

## PROJECTS

**Distributed Agent Execution Platform** | *Go, PostgreSQL, Docker, Kubernetes, Gemini API*

- Built a distributed agent execution platform with a Go API, concurrent Go workers, and PostgreSQL-backed durable queueing, integrating Gemini and validated read-only tools for asynchronous incident investigations.
- Implemented idempotent submissions, renewable leases, and ownership tokens for bounded crash recovery and stale-result rejection; tested worker termination, API restarts, and database outages.
- Deployed containerized services on Kubernetes with persistent storage, health probes, resource limits, and graceful shutdown; validated with integration, failure-injection, and race-checked Go tests.

**Registration Notifier** | *JavaScript, HTML/CSS, Chrome Extension API*

- Developed a Chrome extension to monitor university course registration, featuring a custom UI for users to manage watchlists, toggle background polling or sound, and receive real-time alerts when targeted classes open.
- Engineered a robust background scraper to reliably parse HTML, extracting and cleaning course metadata.
- Implemented Manifest V3 architecture with Service Workers for polling, Chrome Local Storage for state management, and Offscreen Documents for audio playback.

**Memory Manager** | *C, GDB, Linux*

- Built a dynamic memory allocator in C, replicating the core functionality of malloc, free, and realloc.
- Designed a segregated free list with a best-fit policy to optimize for speed and reduce memory fragmentation.
- Ensured efficient memory utilization by coalescing and splitting free blocks to satisfy allocation requests.